Sandip Foundation’s
Sandip Institute of Engineering and Management, Nashik
Department of Electronics and Telecommunication Engineering
Academic Year: 2024 - 2025

Advanced Assistive Brain Controlled Wheelchair For Physically Disabled Person’s

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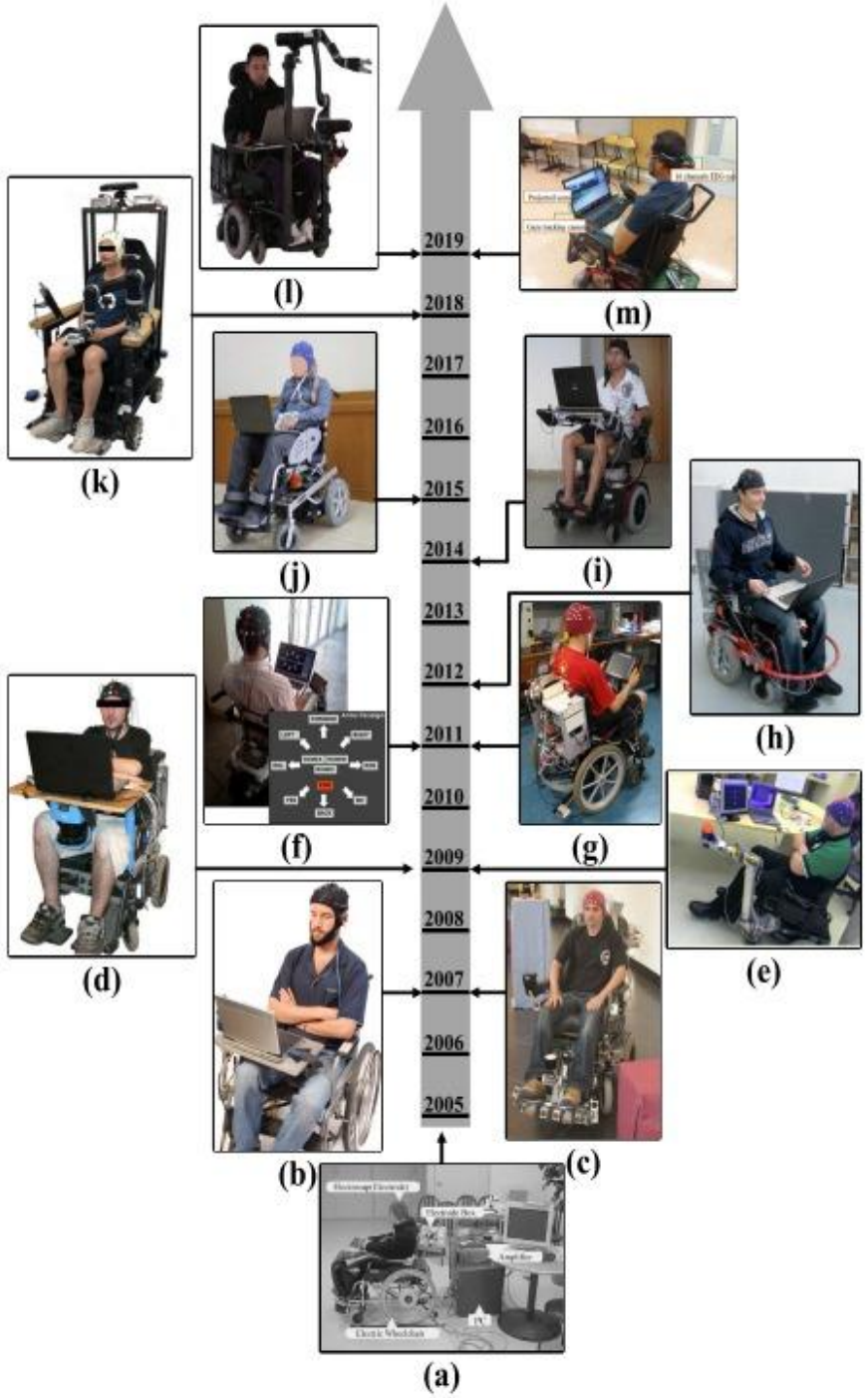
Introduction

- Brain is one of the human body's biggest and most complex organs. It consists of over 100 billion nerves that interact in trillions of synapses-called links. The brain consists of many specialised fields which work together.
- Many people are suffering from the quadriplegia around the world which affects their four limbs.
- The Intelligent Brain-Controlled Wheelchair System (IBCMS) is an innovative assistive technology project designed to empower individuals with paralysis or severe mobility impairments. This comprehensive system integrates brain-control interfaces, health monitoring, obstacle avoidance, GPS/GSM tracking, and home automation to enhance mobility, independence, and quality of life. Utilizing Eye Tracking and Voice Control methods, the IBCMS enables users to navigate their surroundings effortlessly. Additionally, the Paralysis Health Monitor tracks vital signs, including oxygen levels, heart rate, body temperature, and blood pressure, ensuring timely medical interventions. Obstacle avoidance and collision detection systems ensure safe navigation, while GPS and GSM tracking enable real-time location monitoring. The system incorporates home automation features, allowing users to control lighting, temperature, doors, and appliances seamlessly. The project aims to significantly improve the lives of individuals with paralysis, enhancing their autonomy, dignity, and overall well-being.

Literature Survey

Technologies Used: Brain to Computer Interface (BCI Technique)

1. EEG Based System: Using EEG Sensors to Acquire Brain Signals.
2. EcoG Based System: Sensors Directly On the Brain Surface to Acquire high Resolution Signals
3. FNIRS Based System: To measure the changes in blood oxygen level to detect emotion of patients.



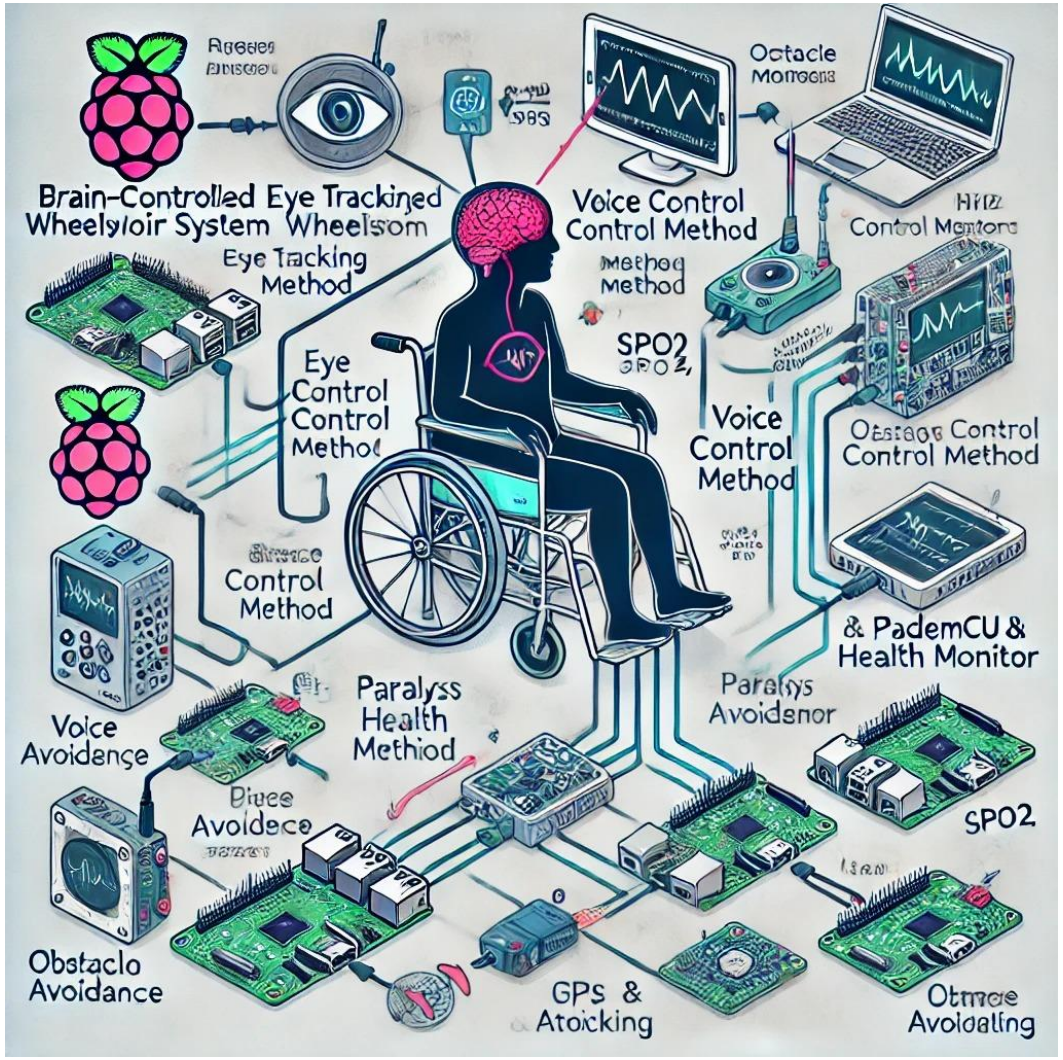
Motivation

- This innovative project aims to empower individuals with paralysis or mobility impairments by developing a cutting-edge, Raspberry Pi-based Brain-Controlled Wheelchair. Integrating Eye Tracking, Voice Control, and advanced health monitoring (oxygen, SPo2, heart rate, body temperature) features, this wheelchair ensures users' safety and well-being. Additionally, Obstacle Avoidance, GPS/GSM tracking, and Home Automation capabilities enhance independence and convenience. By harnessing the latest technologies, this project seeks to transform lives and promote inclusivity for people with disabilities.

MAIN OBJECTIVE

- Design a system to control wheelchair through brain.
- Design a system to control Home Automation appliances such as Fan , light etc through brain with efficiency of 100%
- The world population is increase faster and also the people disabilities (PWD) ratio is very Fastly increasing.
- To create a smart wheelchair that uses eye-tracking technology to allow individuals with severe physical disabilities to navigate their environment independently, safely, and comfortably, while also providing additional features that enhance their overall quality of life.
- The wheelchair integrates eye-tracking, voice control, obstacle avoidance, and GSM & GPS technologies to enhance mobility, safety, and communication.

Structure of the System



- **Brain-controlled:** User Can operate the Wheelchair through EEG Signals.
- **Eye tracking:** The user can control the wheelchair with their eyes.
- **Voice control:** The user can control the wheelchair with their voice.
- **Obstacle avoidance:** The wheelchair can automatically avoid obstacles.
- **Paralysis health monitoring:** The wheelchair can monitor the user's health.
- **GPS & auto-tracking:** The user can track their location and receive directions.
- **Spo2, Temperature and Pulse Rate:** The wheelchair can monitor the user's blood oxygen levels, Temperature and pulse

Hardware Requirements

- Rasbeery Pi 4
- Arduino
- NodeMcu ESP8266
- Neurosky Sensor
- Web-cam
- IR Sensor
- LN298N Motor Driver
- DHT11 Temperature Sensor
- Heartbeat Sensor
- Dc Motors
- Power supply
- Neo-6M GPS Module
- SIM800L GSM Module

Software Requirement

- Tinkercad and Proteus 8 (Simulation)
- Rasberry pi OS
- BCI Technique /Algorithm
- Tensor Flow or Keras (Voice)
- Py Serial lib (For GPS)
- Py Serial lib for Health Monitoring

Proposed System

- In the Brain Control Wheelchair we have to add a system that controls appliances using brainwaves. Neurosky sensors on the head send signals via Wi-Fi to a Node MCU (a microcontroller), which is connected to the appliance. The system is programmed so that brain activity triggers corresponding actions on the appliance (e.g., thinking of closing a fan causes the fan to close). The diagram illustrates the flow of information from the brain to the appliance through the hardware platform

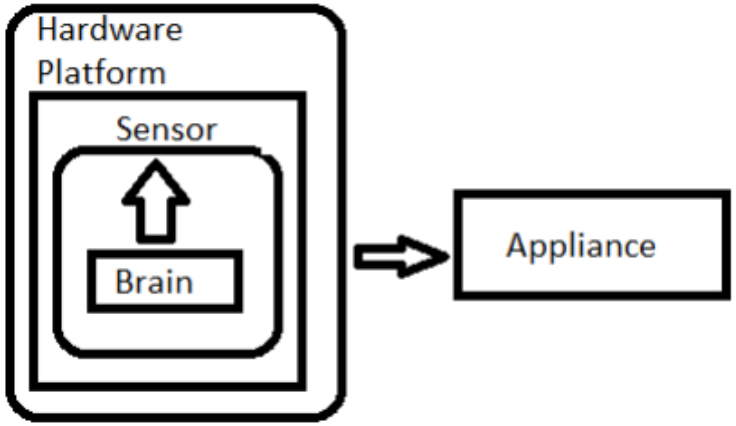


Fig 4 (a): Block Diagram "Control Device through Brain "

- A test or record of brain activity produced by brain is called EEG (Electroencephalogram). Sensor through which Brain signal will be read is Neurosky.

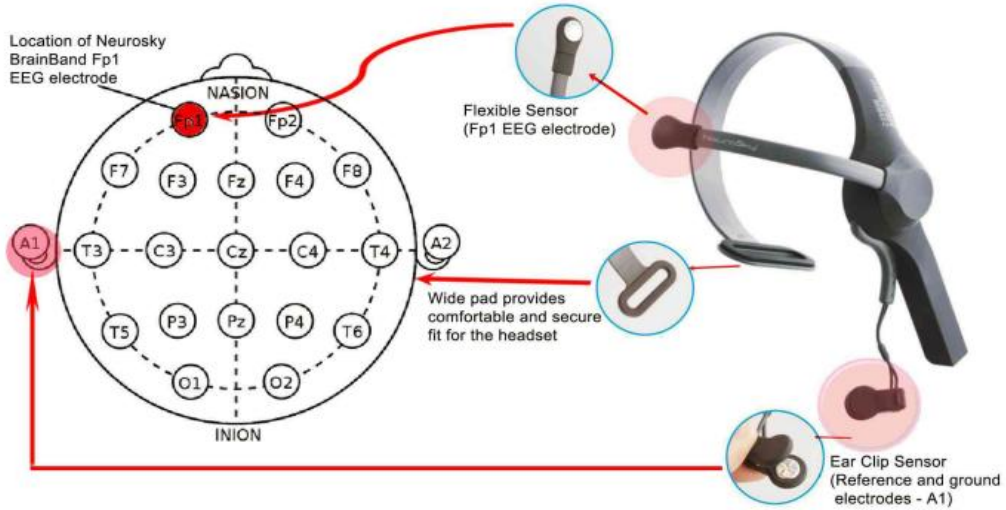
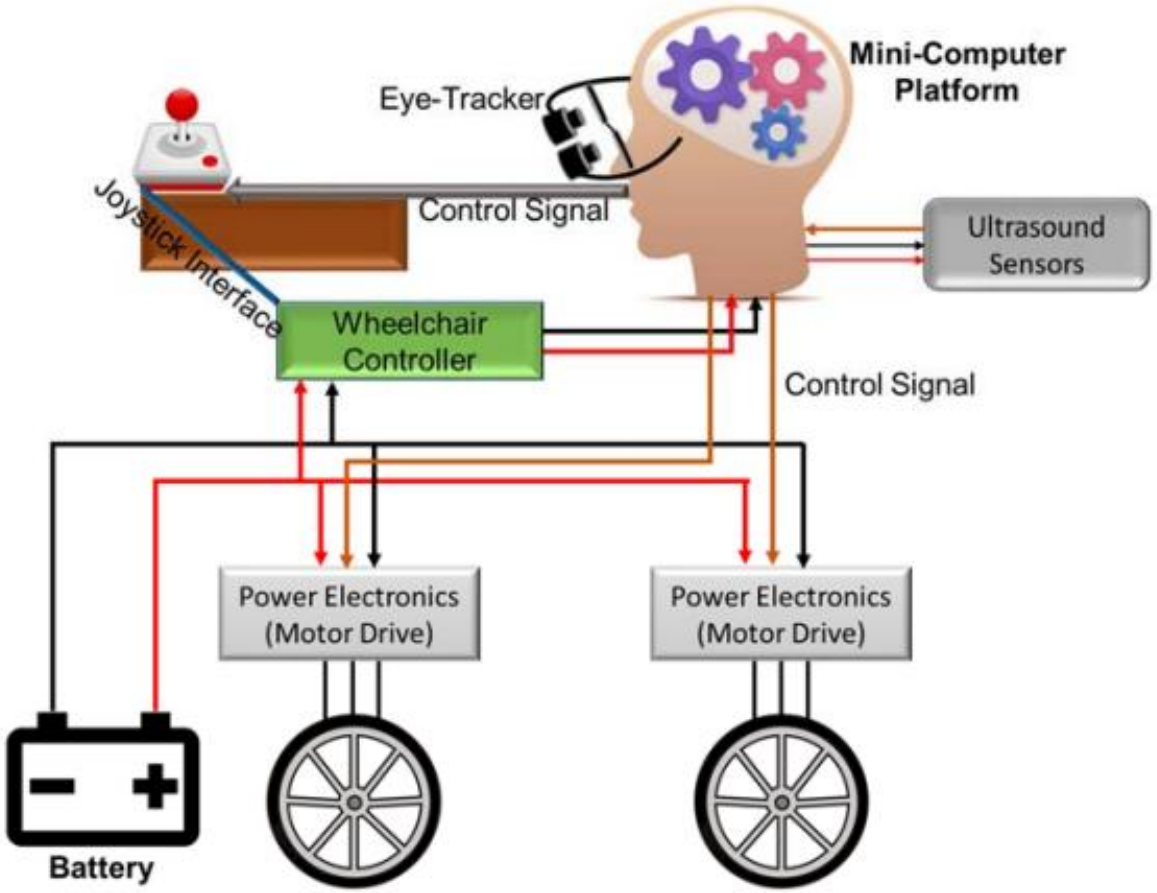


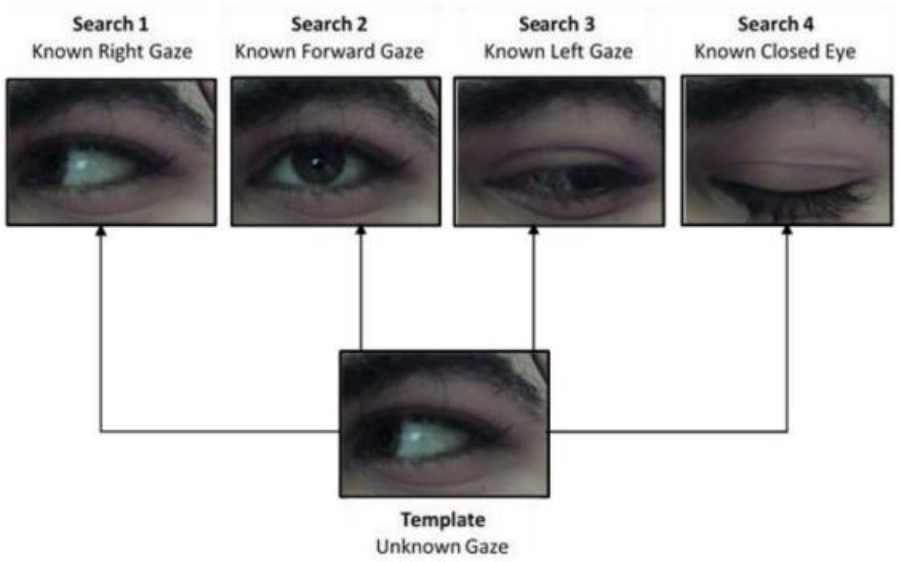
Fig 5: Neurosky Brain Sensor

- The Brain-computer interface (BCI) system using EEG to control a wheelchair. An EEG headset (Neurosky) reads brainwave activity, which is transmitted via Bluetooth to STM32. The STM32 processes the data and sends signals to DC motor drivers to control the wheelchair's movement. An Android application likely provides monitoring and control options

Additional System



Gaze Estimation

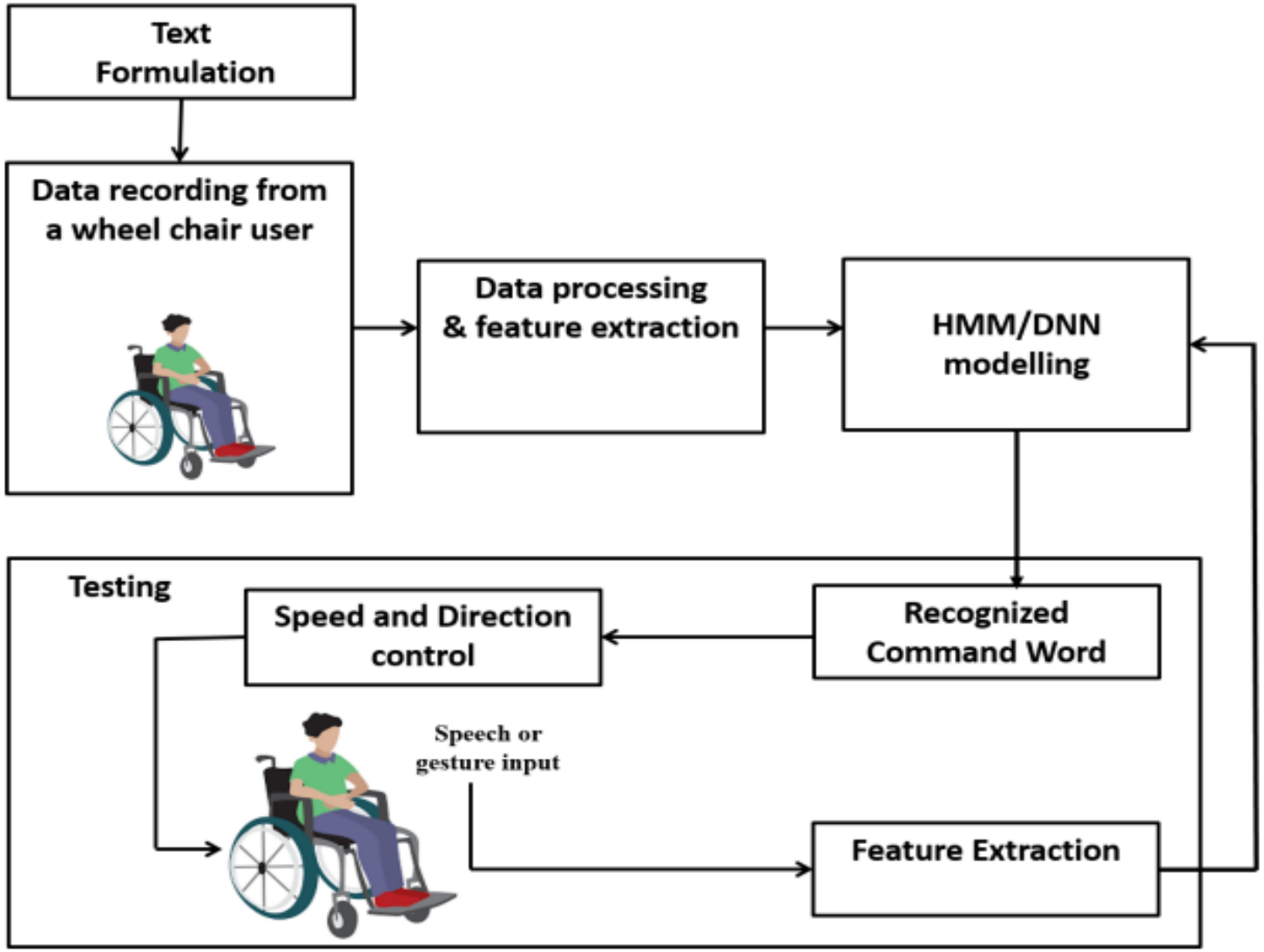


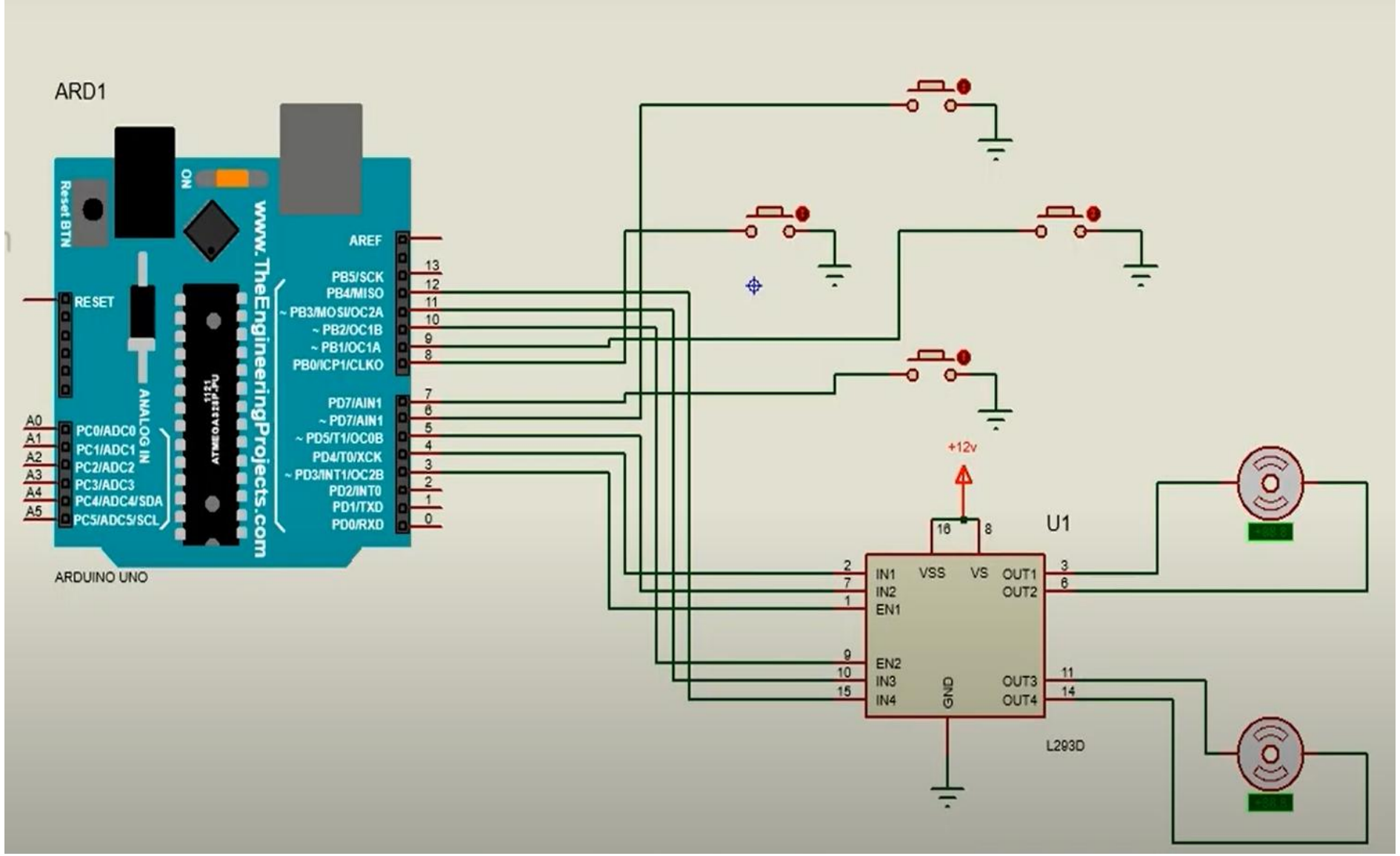
The image depicts a gaze estimation process by comparing an unknown gaze image with a set of known gaze templates. The known templates include right gaze, forward gaze, left gaze, and closed eye. By matching the unknown gaze against these reference images, the system can classify the direction or state of the gaze. This approach, known as template matching, helps determine where a person is looking or if their eyes are closed by finding the closest match between the unknown gaze and the predefined templates.

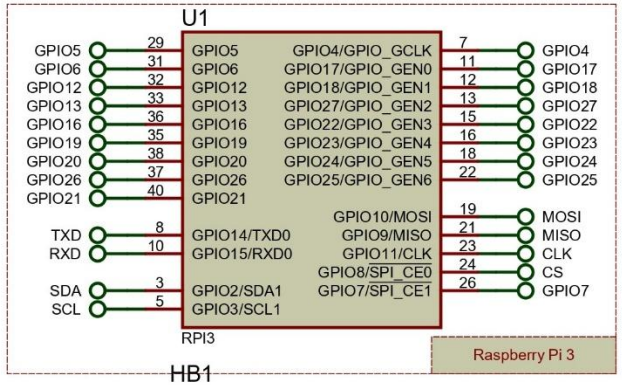
Advantages

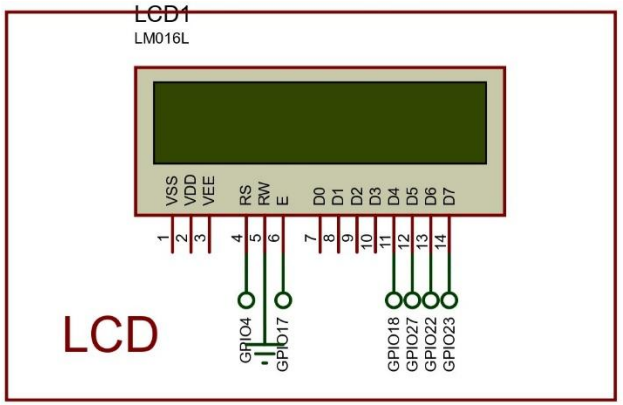
- Increased Independence: Users can control their wheelchair without support of others to our Brain
- Hands Free Control: Allowing Users with Serve Physical Disabilities to operate the wheelchair Using Eye Moment
- Enhancing Independence: Provide and accessible way for users to control the wheelchair using Voice Command
- Real Time Health Monitoring: Ensuring the User’s Health is tracked Real time and Alerts Caregivers to any Abnormalities.
- Ensure Safety Of Patients: Its Automatically Detecting the Obstacles and Preventing Injuries.
- Real time Location Tracking: Allows Location tracking that Enhance User Safety.

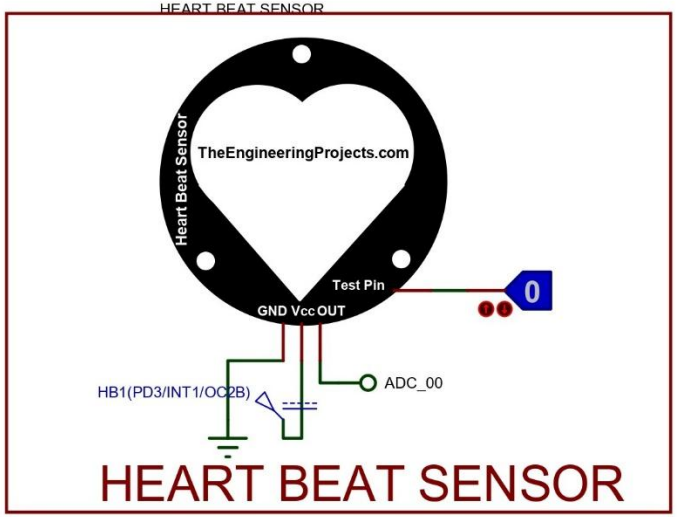
Voice Controlled

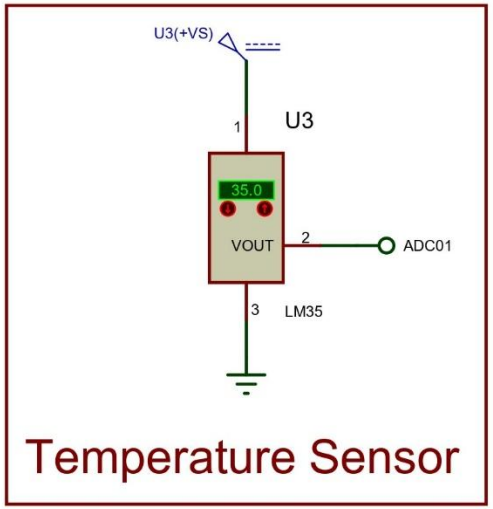


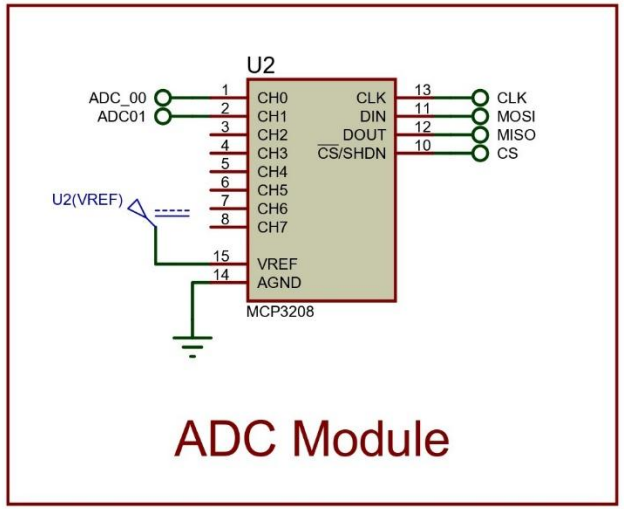

ARD1
Reset BTN
ON
RESET
ANALOG IN
A0
A1
A2
A3
A4
A5
PC0/ADC0
PC1/ADC1
PC2/ADC2
PC3/ADC3
PC4/ADC4/SDA
PC5/ADC5/SCL
www.TheEngineeringProjects.com
AREF
PB5/SCK
PB4/MISO
~ PB3/MOSI/OC2A
~ PB2/OC1B
~ PB1/OC1A
PB0/ICP1/CLKO
PD7/AIN1
~ PD7/AIN1
~ PD5/T1/OC0B
PD4/T0/XCK
~ PD3/INT1/OC2B
PD2/INT0
PD1/TXD
PD0/RXD
ARDUINO UNO
+12v
U1
IN1
IN2
EN1
EN2
IN3
IN4
VSS
VS
GND
OUT1
OUT2
OUT3
OUT4
L293D

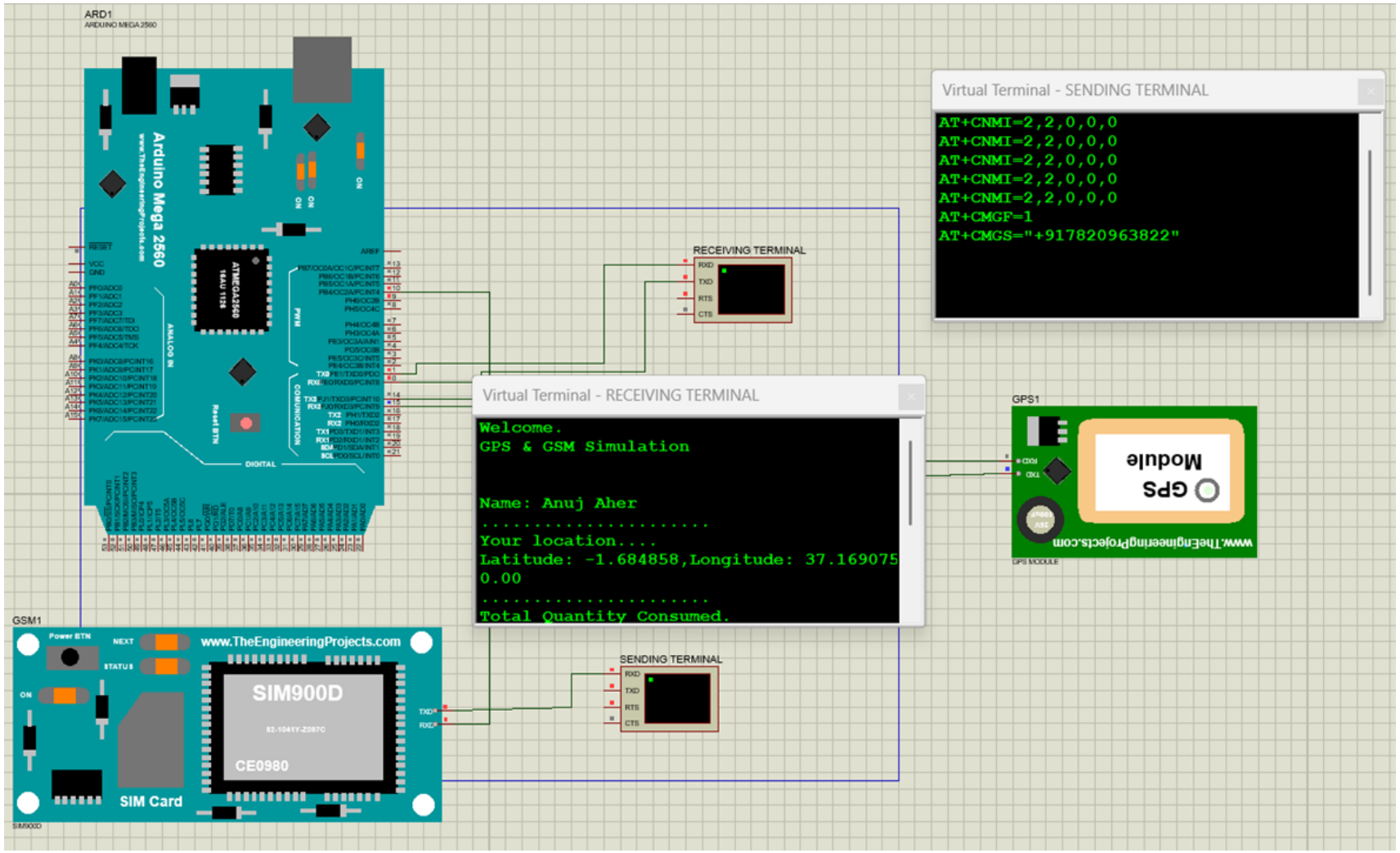

U1
GPIO5
GPIO6
GPIO12
GPIO13
GPIO16
GPIO19
GPIO20
GPIO26
GPIO21
TXD
RXD
SDA
SCL
GPIO4/GPIO_GCLK
GPIO17/GPIO_GEN0
GPIO18/GPIO_GEN1
GPIO27/GPIO_GEN2
GPIO22/GPIO_GEN3
GPIO23/GPIO_GEN4
GPIO24/GPIO_GEN5
GPIO25/GPIO_GEN6
GPIO10/MOSI
GPIO9/MISO
GPIO11/CLK
GPIO8/SPI_CE0
GPIO7/SPI_CE1
GPIO14/TXD0
GPIO15/RXD0
GPIO2/SDA1
GPIO3/SCL1
MOSI
MISO
CLK
CS
RPI3
Raspberry Pi 3


LCD1
LM016L
VSS
VDD
VEE
RS
RW
E
D0
D1
D2
D3
D4
D5
D6
D7
GPIO4
GPIO17
GPIO18
GPIO27
GPIO22
GPIO23
LCD

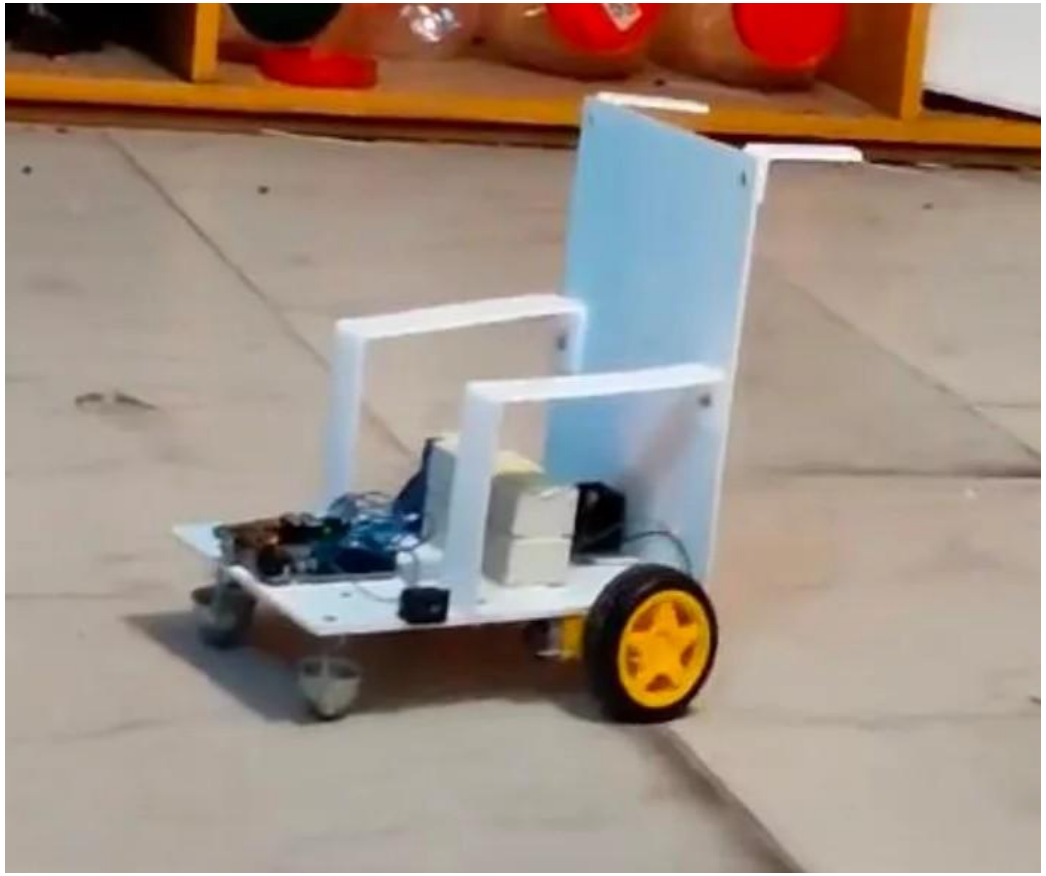

HB1
HEART BEAT SENSOR
Heart Beat Sensor
TheEngineeringProjects.com
Test Pin
GND Vcc OUT
0
HB1(PD3/INT1/OC2B)
ADC_00
HEART BEAT SENSOR

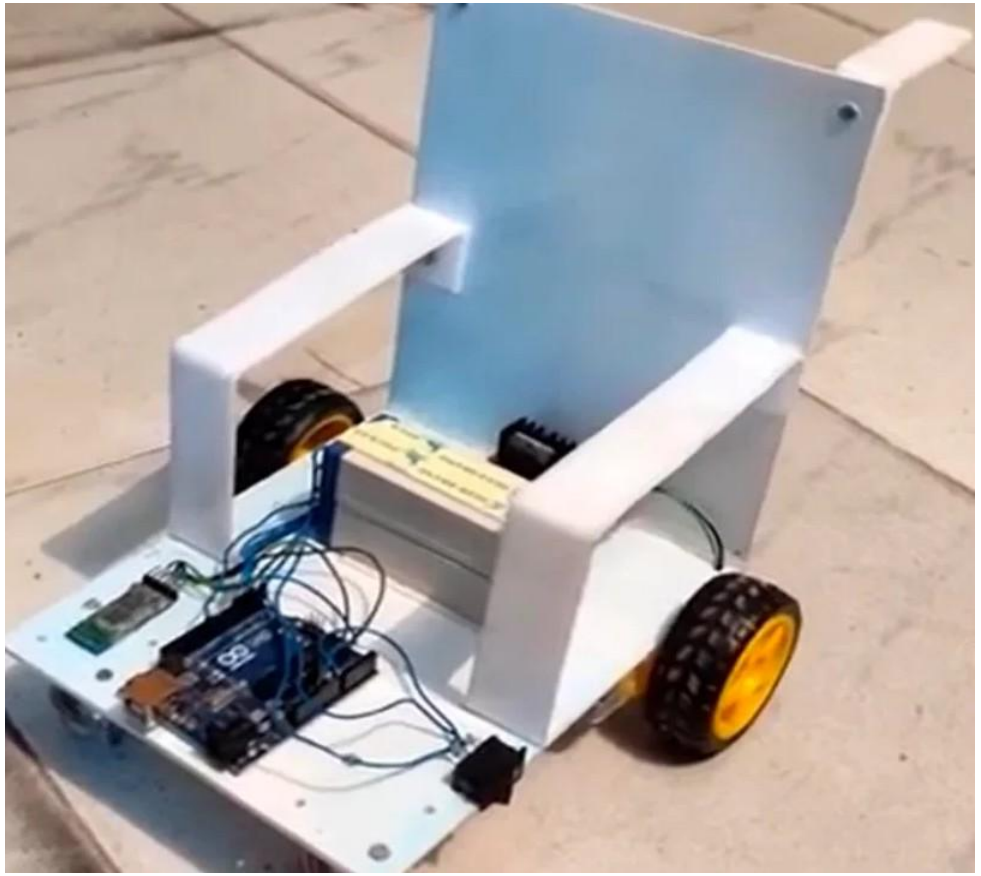

U3(+VS)
U3
35.0
VOUT
ADC01
LM35
Temperature Sensor


U2
ADC_00
ADC01
CH0
CH1
CH2
CH3
CH4
CH5
CH6
CH7
CLK
DIN
DOUT
CS/SHDN
CLK
MOSI
MISO
CS
U2(VREF)
VREF
AGND
MCP3208
ADC Module

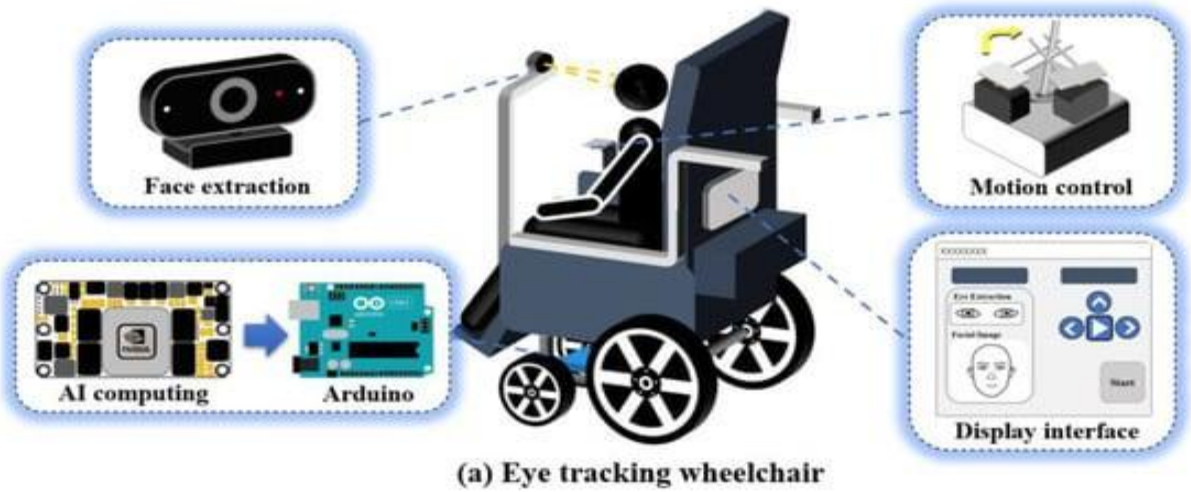

ARD1
ARDUINO MEGA 2560
Arduino Mega 2560
ATMEGA2560
RECEIVING TERMINAL
RXD
TXD
RTS
CTS
Virtual Terminal - SENDING TERMINAL
AT+CNMI=2,2,0,0,0
AT+CNMI=2,2,0,0,0
AT+CNMI=2,2,0,0,0
AT+CNMI=2,2,0,0,0
AT+CNMI=2,2,0,0,0
AT+CMGF=1
AT+CMGS="+917820963822"
Virtual Terminal - RECEIVING TERMINAL
Welcome.
GPS & GSM Simulation
Name: Anuj Aher
.....
Your location....
Latitude: -1.684858,Longitude: 37.169075
0.00
.....
Total Quantity Consumed.
GPS1
GPS Module
www.TheEngineeringProjects.com
GPS MODULE
GSM1
Power BTN
NEXT
STATUS
ON
www.TheEngineeringProjects.com
SIM900D
CE0980
SIM Card
SENDING TERMINAL
RXD
TXD
RTS
CTS

Working Model

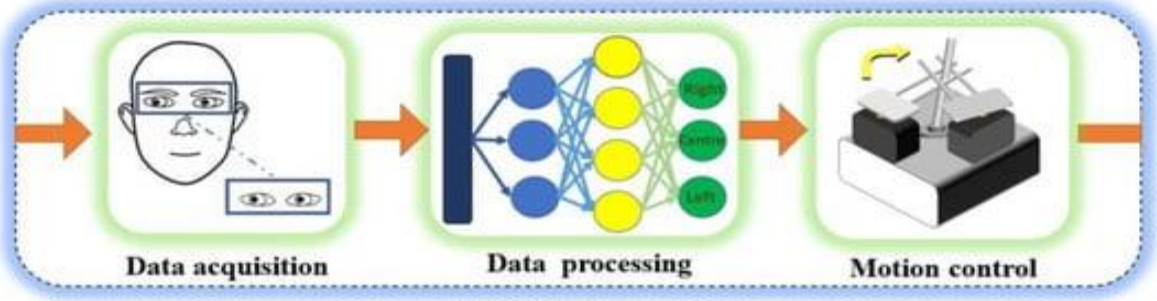




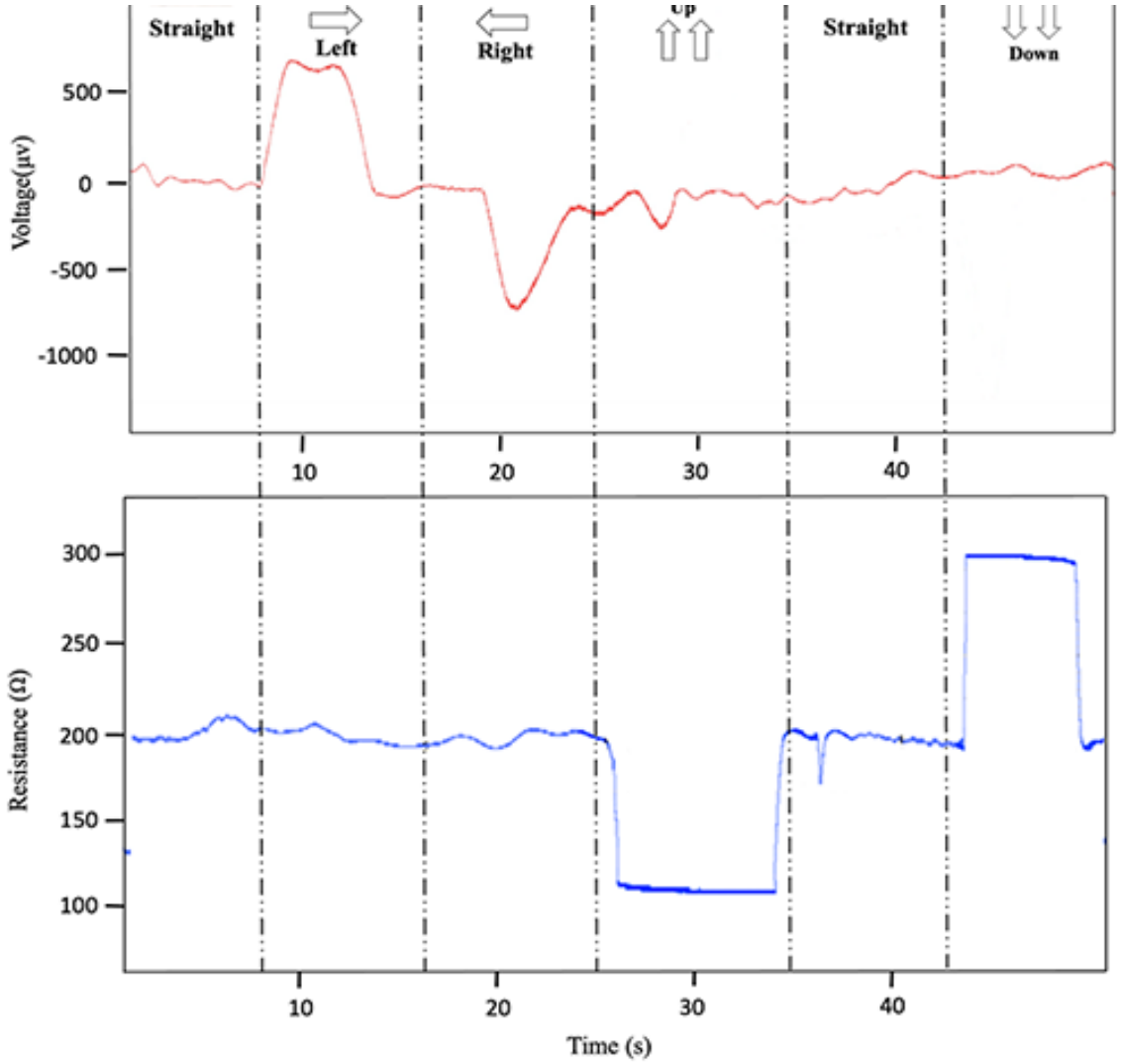
Eye Controlled

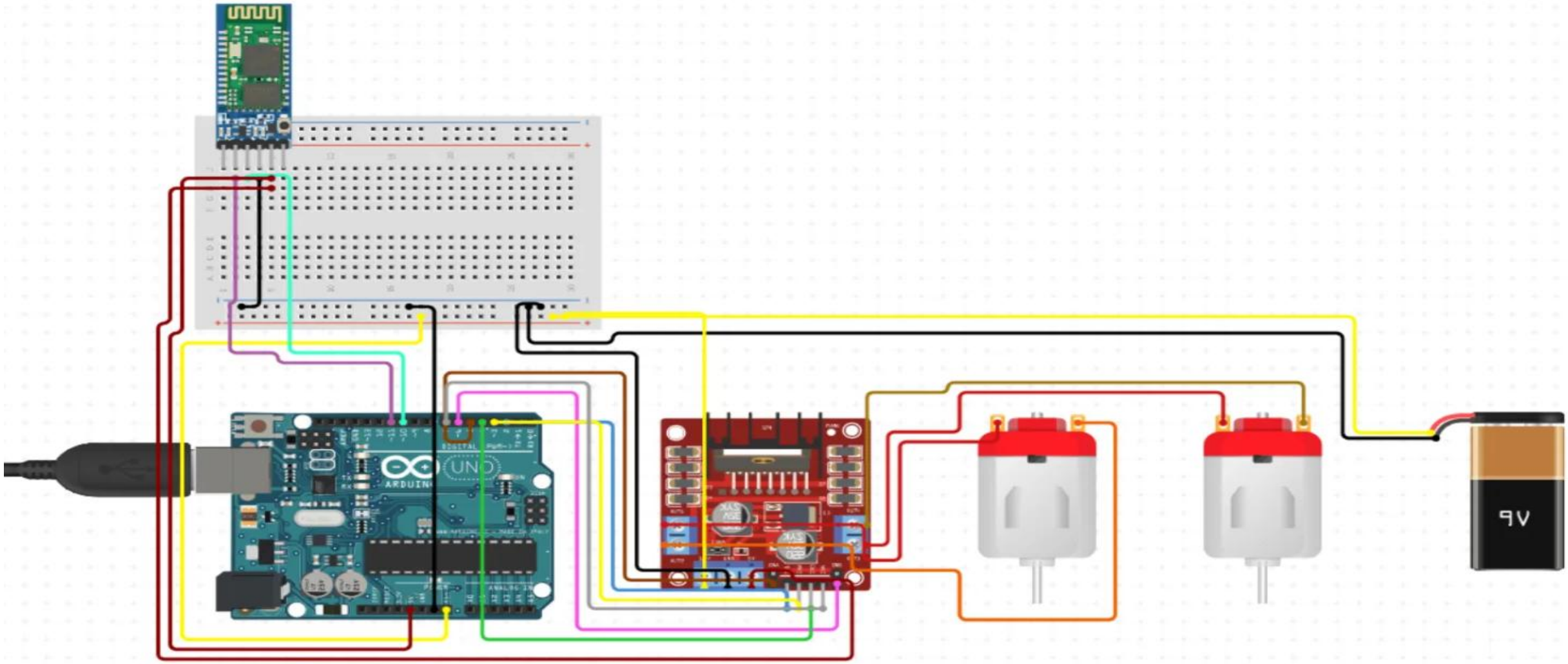


(a) Eye tracking wheelchair

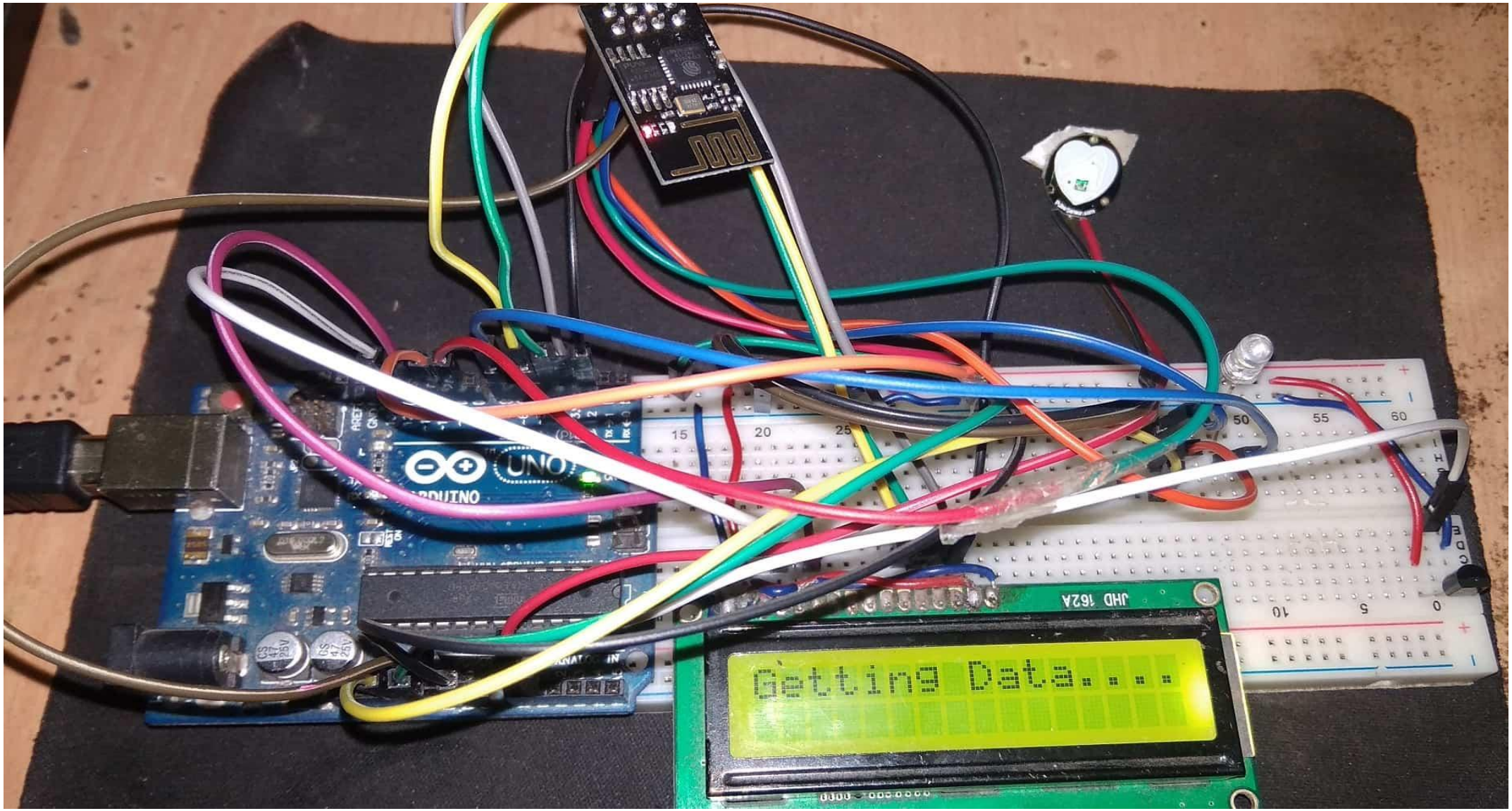


(b) Control flowchart

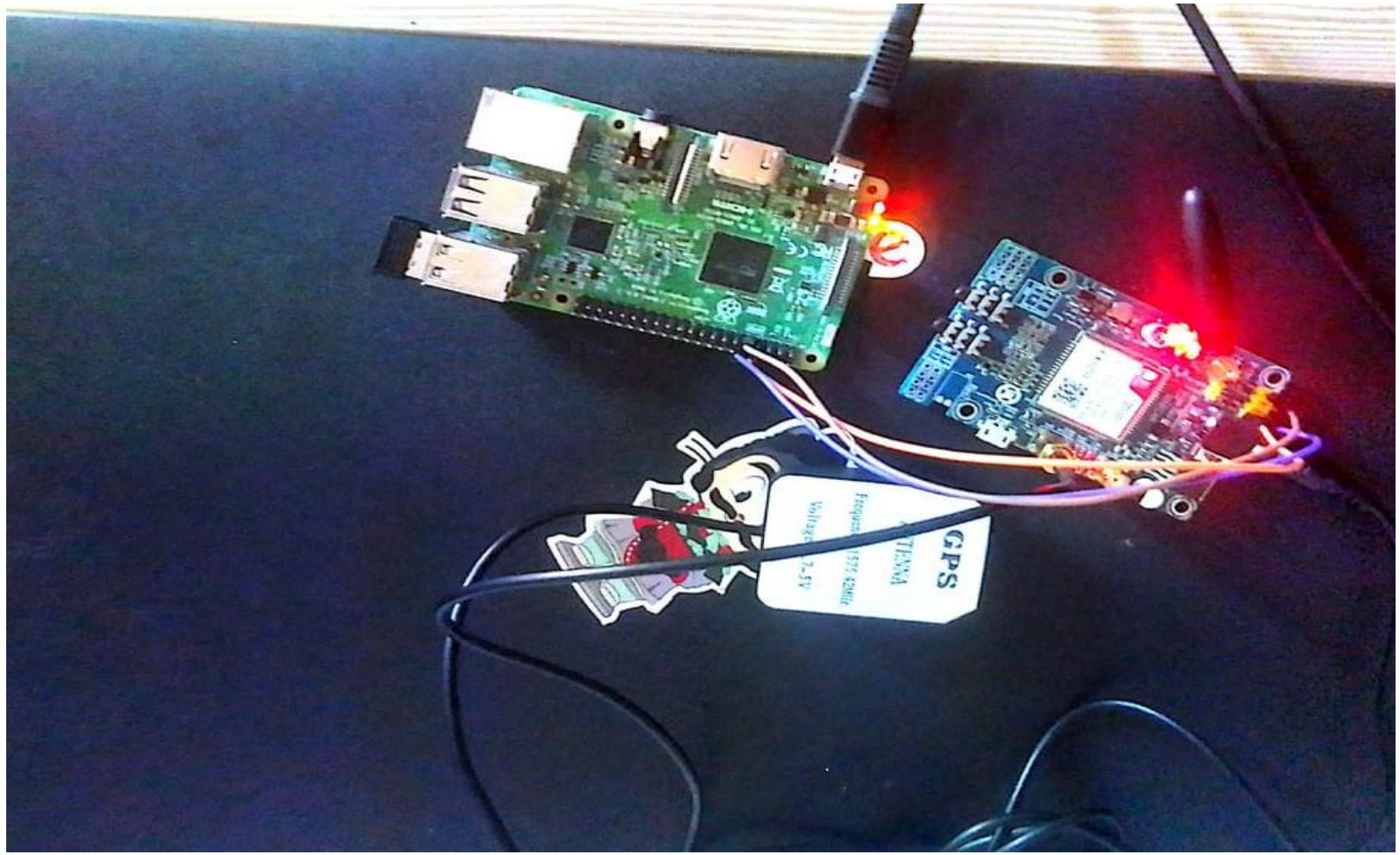


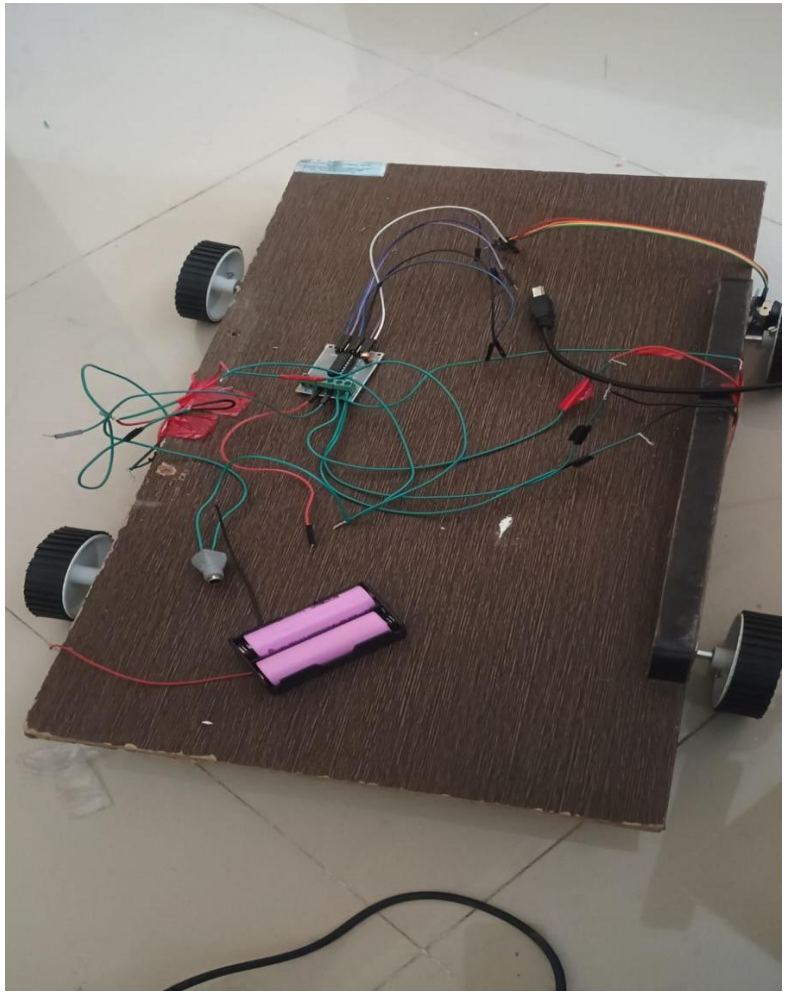

UNO
ARDUINO
9V

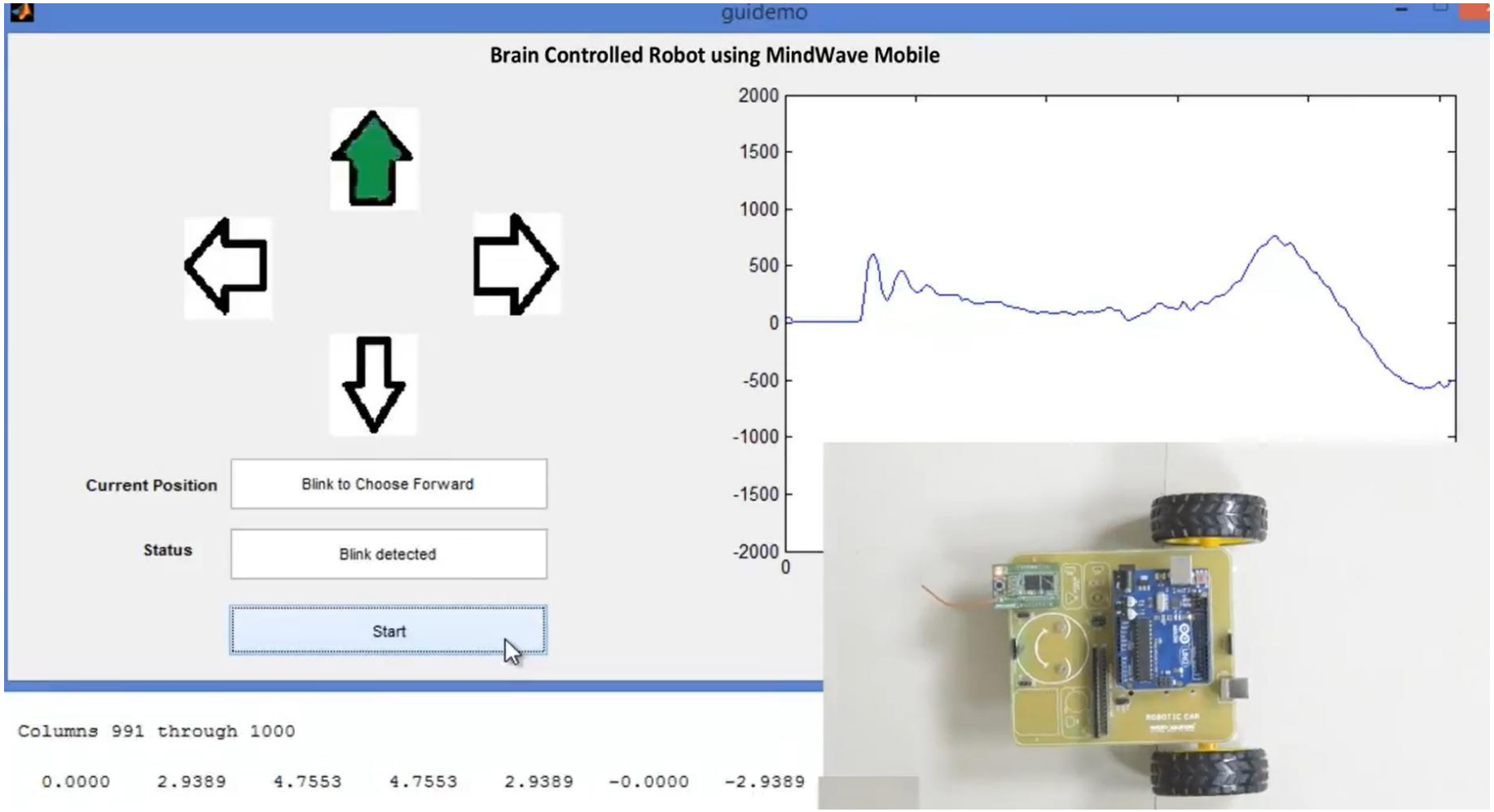
Health Monitoring Testing

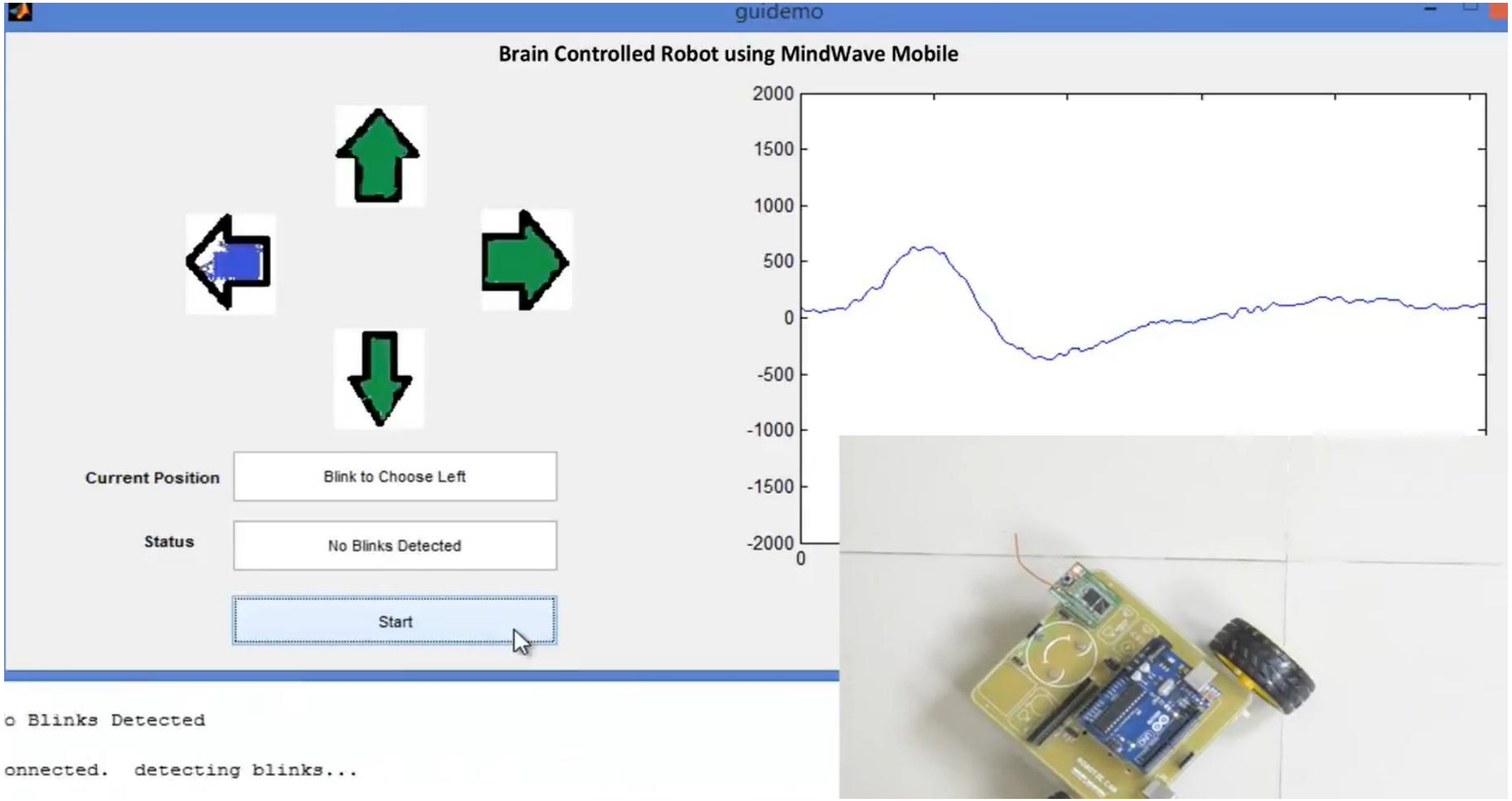


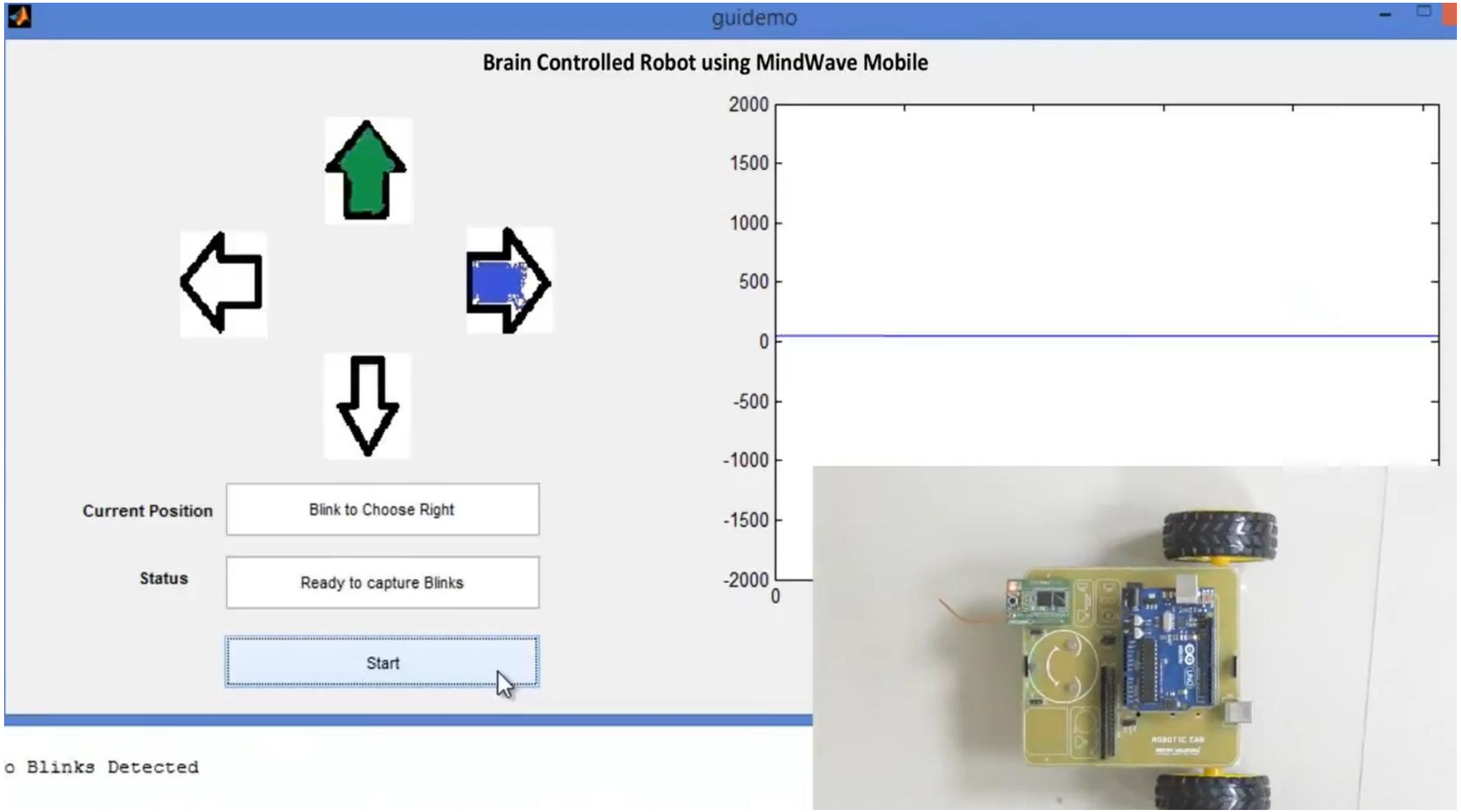
GPS and GSM Testing






guidemo
Brain Controlled Robot using MindWave Mobile
Current Position
Blink to Choose Forward
Status
Blink detected
Start
2000
1500
1000
500
0
-500
-1000
-1500
-2000
0
Columns 991 through 1000
0.0000 2.9389 4.7553 4.7553 2.9389 -0.0000 -2.9389


guidemo
Brain Controlled Robot using MindWave Mobile
2000
1500
1000
500
0
-500
-1000
-1500
-2000
0
Current Position
Blink to Choose Left
Status
No Blinks Detected
Start
o Blinks Detected
onnected. detecting blinks...



o Blinks Detected

Future Scope

- The Raspberry Pi Brain-Controlled Wheelchair project has vast potential for future advancements. Future developments may include integrating advanced sensors, artificial intelligence, and machine learning algorithms for enhanced user experience and predictive health monitoring. Expansion plans could involve commercialization, customizable designs, affordability, and collaboration with rehabilitation centers. Incorporating emerging technologies like IOT, 5G connectivity, augmented reality, and block chain-based security may also enhance the project's functionality and security. Further research and clinical trials will validate the project's effectiveness, guiding continuous improvement and innovation.

Conclusion

- The Raspberry Pi Brain-Controlled Wheelchair project revolutionizes mobility assistance for individuals with paralysis or severe impairments. By integrating eye-tracking, voice control, and paralysis health monitoring (oxygen, SPo2, heartbeat, body temperature), this innovative solution enhances user autonomy, safety, and quality of life. Obstacle avoidance, GPS/GSM tracking, and home automation features further ensure seamless navigation and independence. This cost-effective, customizable, and scalable project bridges engineering, medicine, and rehabilitation, offering vast potential for future advancements and improved accessibility.This innovative solution demonstrates the potential to improve the quality of life for those with severe disabilities, offering increased independence and security.

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- https://ieeexplore.ieee.org/document/8243755
- https://www.sciencedirect.com/science/article/abs/pii/S221478532200760X
- https://www.journal-dogorangsang.in/no_2_Online_22/5_april.pdf

- https://www.journal-dogorangsang.in/no_2_Online_22/5_april.pdf